

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
 Nov – Dec 2019

Max. Marks: 100

Class: SY

Name of the Course: Database Management Systems

Course Code: 2UIC303

Duration: 3 Hrs

Semester: III

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q1 (a)	What are the features of database system? Give advantages and disadvantages of database system.	10
Q1 (b)	What is the difference between logical data independence and physical data independence? OR What is Database Schema? Explain different types of schema and their relationship using appropriate example.	10
Q 2 (a)	Design an ER Diagram for a BANKING SYSTEM that includes the following concepts: a. Specialization b. Generalization c. Cardinality d. Relationship e. Entity	10
Q. 2 (b)	Write SQL Query for the following set of tables: EMPLOYEE (<u>EmpNo</u> , DateOfBirth, Name, Address, Gender, Salary, Dnumber) Department (<u>DNo</u> , Dname, ManagerEmpNo, ManagerStartDate) a. Display the age of 'male' employees. b. Display all employees in department 'Marketing'. c. Display the name of highest salary paid 'female' employee. d. Which employee is oldest manager in company. e. Display the name of department of the employee 'SMITH'.	10

	OR Write the relational algebra expression for the following query. The structure of the database is given as below: S (Supplier Table) { S#, SNAME, STATUS, CITY} P (Product Table) { P#, PNAME, COLOR, WEIGHT} SP ({ S#, P#, QTY} - Get full details of all products of "Red" Color. - Get supplier name and product name for the shipment where the quantity is in the range 300 to 400. - Get supplier number for the supplier who supply product "P1". - Get the name of the product supplied by the supplier of city "Mumbai". - Get the total quantity of the product "Bolt" supplied by supplier "Jack".	
Q 3 (a)	Explain 3 NF and BCNF with respect to different anomalies with example. OR Why decomposition is required during normalization? What are the properties of decomposition? Explain in detail with the help of examples.	8
Q. 3 (b)	Consider the following database of EMPLOYEE EMPLOYEE (<u>EmpId</u>, Adhaar No, EmpName, City, Pincode, Dept Id, Department Name). Normalize the above relation up to 3NF. Give functional dependencies and all assumptions very clearly.	12
Q4 (a)	Why indexing is required? Give the difference between single level and multilevel indexing with example. OR How hashing can be used in Indexing? Take an example of College Database and implement it.	10
Q4 (b)	Explain different phases of query processing with the diagram.	10
Q5 (a)	Write short notes on (any 2) : - Properties of transaction - Log based Recovery - Shadow paging	10
Q.5 (b)	Define serializability. State two phase locking protocol and its advantages and disadvantages.	10

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End Semester Exam

Nov - Dec 2019

Max. Marks:50

Class: **S.Y. B.Tech**

Name of the Course: Database Management Systems - Section A

Branch: IT

Course Code: **UJTC 302**

Duration: 2 Hours

Semester: III

PWD

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q1 (a)	Describe and demonstrate with suitable examples (any two), i. Conceptual schema ii. Physical schema iii. External Schema	5 5
Q1 (b)	Solve any two. i. Demonstrate the relational model with suitable examples ii. Demonstrate the concept of Data independence with suitable example iii. Discuss the roles of database administrator	5 5
Q2 (a)	Discuss the following terms from relational database model with suitable examples (any two). i. Binary relationship ii. Ternary Relationship iii. Participation constraints	5 5
Q2 (b)	Solve any two. i. Apply domain integrity constraints and formulate any two meaningful SQL statements. ii. Apply referential integrity constraints and formulate any two meaningful SQL statements. iii. Apply conditional clauses and formulate any two meaningful SQL statements.	5 5

Q3 (a)	Match the following: <table><tr><td>i. Cardinality</td><td>i. Unique data values</td></tr><tr><td>ii. Entity</td><td>ii. Relationship</td></tr><tr><td>iii. The smallest unit of information about a record</td><td>iii. Attribute</td></tr><tr><td>iv. Object based logical model</td><td>iv. Table</td></tr><tr><td>v. ER diagram diamond</td><td>v. ER model</td></tr></table>	i. Cardinality	i. Unique data values	ii. Entity	ii. Relationship	iii. The smallest unit of information about a record	iii. Attribute	iv. Object based logical model	iv. Table	v. ER diagram diamond	v. ER model	5
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ii. Entity	ii. Relationship											
iii. The smallest unit of information about a record	iii. Attribute											
iv. Object based logical model	iv. Table											
v. ER diagram diamond	v. ER model											
Q3 (b)	Select the correction option: i. SQL command to change the a column length from an existing table a. Drop column b. Alter table c. Update table d. Alter column ii. An attribute or set of attributes within one relation that matches the candidate key of some other relation: a. Super key b. Candidate key c. Foreign key d. Primary key iii. The normalization of 1NF relations to 2NF involves: a. Removal of partial dependency b. Removal of full dependency c. Removal of transitivity d. Removal of multivalued dependency iv. Why to normalize the data? a. To avoid the repetition b. To prevent the fragmentation c. Avoid consistency d. None of above v. The set of permitted values for each attribute is called its a. Attribute sets b. Attribute range c. Domain d. Group	5										

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End Semester Exam

Nov - Dec 2019

Max. Marks:50

Class: **S.Y. B.Tech**

Name of the Course: Database Management Systems - Section B

Branch: IT

Course Code: **UITC302**

Duration: 2 Hours

Semester: III

PWD

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q4(a)	i. Describe dense index with suitable example ii. Describe sparse index with suitable example OR i. Discuss the various parameters to be consider while indexing ii. Explain the measures of query costing.	5 5
Q4(b)	i. Discuss conflict Serializability with suitable example ii. Discuss view Serializability with suitable example	5 5
Q5(a)	i. Explain static hashing technique with example. ii. Explain dynamic hashing technique with example. OR i. Demonstrate serial schedule with suitable example. ii. Demonstrate concurrent schedule with suitable example	5 5
Q5(b)	i. Demonstrate the atomicity property of the relational database with suitable example. ii. Demonstrate data integrity property of the relational database with suitable example.	5 5
Q6	Select the correction option: i. Dynamic hashing allows to? a) Accommodate the growth of the database b) Accommodate the shrinkage of the database c) Allows modification of hash function d) All of the mentioned ii. The space overhead in dynamic hashing is _____ than that of static hashing a) More b) Less	10

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- c) Equal
- d) None of the mentioned

iii. Locks placed by command are called _____.

- a) implicit locks
- b) explicit locks
- c) exclusive locks
- d) shared locks

iv. The situation where no transaction can proceed with normal execution is known as _____

- a) Dead block
- b) Deadlock
- c) Execution halt
- d) Abortion

v. I and J are _____ if they are operations by different transactions on the same data item, and at least one of them is a write operation.

- a) Conflicting
- b) Overwriting
- c) Isolated
- d) Durable

vi. A sequence of primitive operations that can be used to evaluate a query are called as _____

- a) Query evaluation algebra
- b) Query evaluation plan
- c) Query evaluation primitive
- d) Query evaluation engine

vii. If a transaction has obtained a _____ lock, it can read but cannot write on the item

- a) Shared mode
- b) Exclusive mode
- c) Read only mode
- d) Write only mode

viii. Which of the following is not a method in deadlock handling

- a) Deadlock prevention
- b) Deadlock detection
- c) Deadlock recovery
- d) Deadlock distribution

ix. The time it takes to find a particular data item is called as _____

- a) Insertion time
- b) Deletion time
- c) Time overhead
- d) Access time

x. Indices with two or more levels are called as?

- a) Multiple Indices
- b) Multilevel indices
- c) Bi- tri- Indices
- d) None of the mentioned